

CHAPTER 7: THE NORMAL DISTRIBUTION

7.1 The Normal Distribution

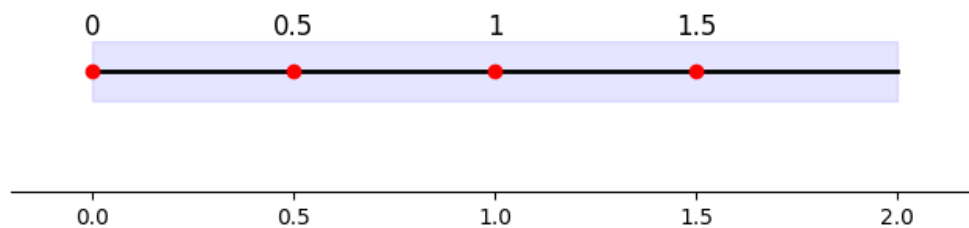
In this section, we explore the central concepts of the normal distribution by first understanding the underlying ideas of continuous random variables and continuous probability distributions. We then delve into the specifics of the normal distribution, its parameters, and its properties.

A. Continuous Random Variable

Definition and Explanation

A **continuous random variable** is one that can take any value within a given interval—or even an unlimited range—unlike a discrete variable, which is restricted to separate, distinct values. Because the set of possible outcomes is uncountably infinite, the probability of the variable taking on any one exact value is zero. Instead, probabilities are assigned to intervals of values.

Continuous Number Line with Sample Points

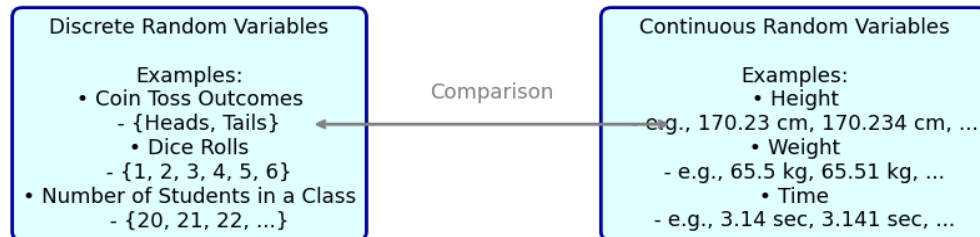


- **Key Points:**
 - **Infinite Possibilities:** The variable can assume any value within the interval (e.g., any real number between 0 and 1, or any height measurement).
 - **Examples:**
 - **Height:** 170.23 cm, 170.234 cm, ...
 - **Weight:** 65.5 kg, 65.51 kg, ...
 - **Temperature:** 22.3°C, 22.3001°C, etc.
 - **Time:** 3.14 seconds, 3.141 seconds, etc.

Example:

Imagine measuring the time it takes for a chemical reaction to complete. Unlike counting the number of leaves on a tree (a discrete countable outcome), time can be any decimal value (e.g., 5.632 seconds). Although you can record time only up to a certain decimal place, theoretically, the value can be infinitely precise.

Diagram 2: Discrete vs Continuous Random Variables



B. Continuous Probability Distribution

Definition and Concept

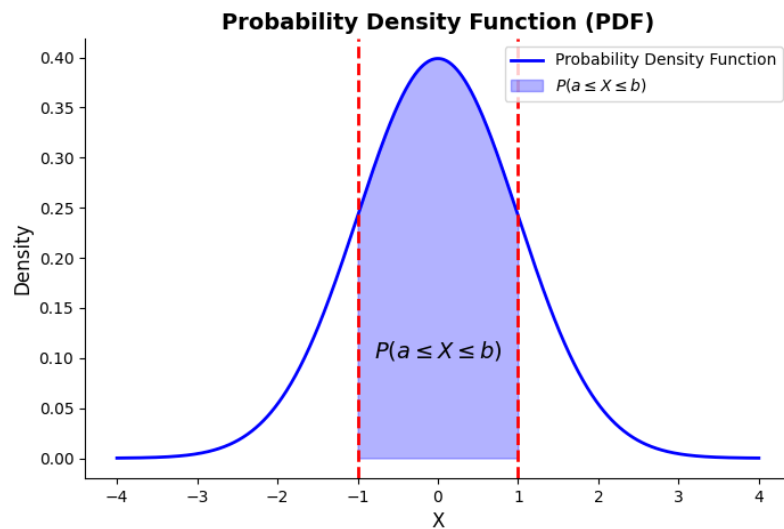
A **continuous probability distribution** describes how the probability is distributed over the values of a continuous random variable. For such variables, the probability of obtaining any specific value is zero. Instead, we define the probability across an interval of values, which is computed as the area under the probability density function (pdf) over that interval.

- **Probability Density Function (pdf):** A function $f(x)$ that describes the likelihood of the random variable taking on a specific value. The probability that the variable X is within an interval $[a, b]$ is given by:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

- **Total Area Under the Curve:** The integral of the pdf over all possible values is equal to 1:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$



This represents the certainty that the variable will assume a value within the defined range.

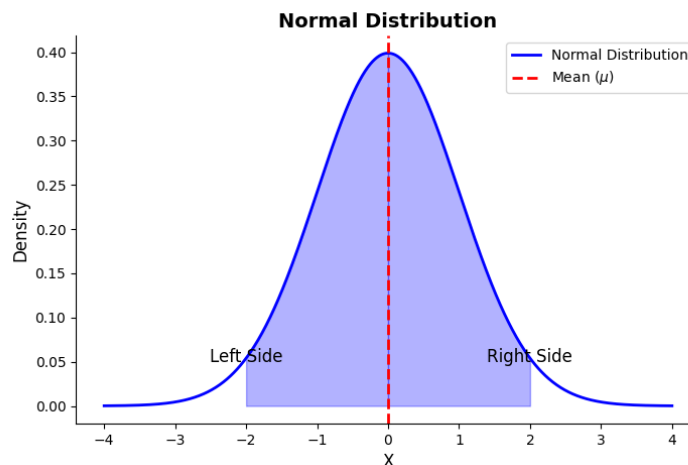
Example:

If X represents the time (in minutes) to complete a task with a continuous probability distribution, the probability that the task takes between 10 and 12 minutes is determined by calculating the area under the pdf from $x = 10$ to $x = 12$.

C. Normal Distribution

The **normal distribution** is arguably the most significant continuous probability distribution in statistics due to its ubiquitous appearance in natural phenomena and measurement errors. It is characterized by its iconic bell-shaped, symmetrical curve.

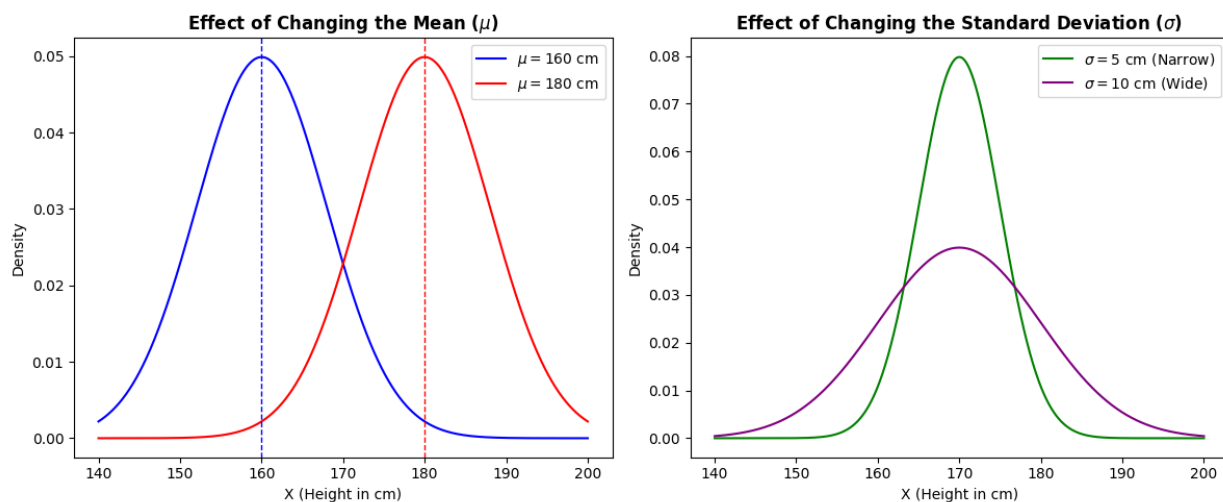
1. Characteristics and Description



- **Bell-Shaped and Symmetrical Curve:** The normal distribution is defined by its smooth, symmetric bell-shaped curve. The symmetry implies that the left half is a mirror image of the right half.
 - **Key Feature:** The distribution is perfectly symmetric about the mean.
- **Mean = Median = Mode:** Because of its symmetry, the central tendency measures—mean, median, and mode—are identical and all located at the center of the distribution.
- **Total Area Under the Curve Equals 1:** The total area under the standard normal curve represents the sum of all probabilities, which is 1.
- **Asymptotic Nature:** The tails of the normal distribution extend infinitely in both directions (positive and negative), approaching the horizontal axis but never touching it. This indicates that every value, however extreme, has a nonzero probability (albeit extremely small).
- **Points of Inflection:** The curve changes its curvature at two points, located one standard deviation from the mean, specifically at $\mu - \sigma$ and $\mu + \sigma$. These are the inflection points where the curvature transitions from concave to convex.
- **Empirical Rule (68–95–99.7 Rule):** This rule provides a quick summary of how data are spread in a normal distribution:
 - Approximately **68%** of the data falls within one standard deviation of the mean.
 - Approximately **95%** of the data falls within two standard deviations.
 - Approximately **99.7%** of the data falls within three standard deviations.

2. Parameters of the Normal Distribution

The exact shape and position of a normal distribution curve are completely determined by two parameters:



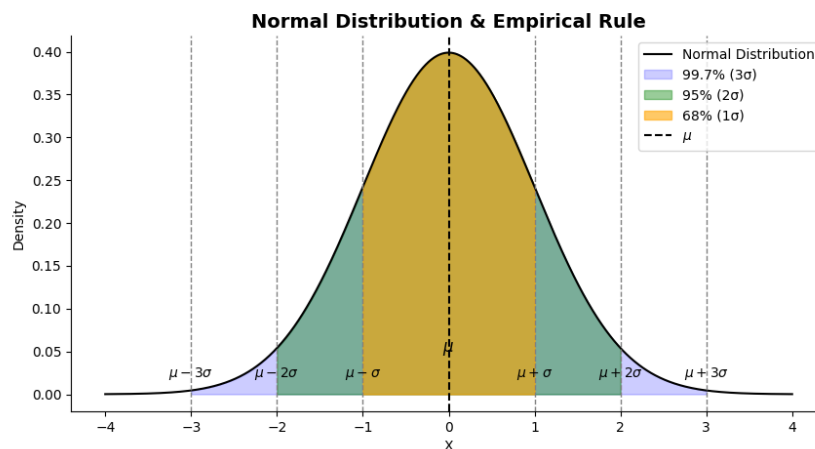
- **Mean (μ):**
 - **Definition:** The mean represents the central or average value of the distribution.
 - **Role in the Distribution:** It determines the center of the bell curve.
 - **Example:** For a dataset representing adult heights, if $\mu = 170$ cm, the peak of the distribution is at 170 cm.

- **Variance (σ^2) and Standard Deviation (σ):**
 - **Variance (σ^2)** measures the spread or dispersion around the mean.
 - **Standard Deviation (σ)** is the square root of the variance and gives a typical measure of deviation from the mean.
 - **Example:** If the variance of exam scores is 25, the standard deviation is $\sigma = 5$. A larger standard deviation implies a wider (flatter) curve, indicating more variability.
- **Notation:** When a random variable X is normally distributed with mean μ and variance σ^2 , we denote it as:

$$X \sim N(\mu, \sigma^2)$$

3. Properties of the Normal Distribution Curve

- **Bell-Shaped and Symmetrical:** The normal curve is smooth, symmetric around the mean μ , and has exactly half its area on each side of the mean.
- **Mean = Median = Mode:** Owing to symmetry, these three measures coincide.
- **Total Area Under the Curve is 1:** This is a requirement of any probability distribution, ensuring that the total probability is 100%.
- **Asymptotic to the x-axis:** The tails extend infinitely toward both positive and negative directions, never touching the x-axis. This implies that no value is impossible, though extreme values have very low probabilities.
- **Points of Inflection:** Occur at $\mu \pm \sigma$; these points mark where the curvature of the graph changes.
- **Empirical Rule (68–95–99.7):**
 - 68% of the values lie within one standard deviation of the mean ($\mu - \sigma$ to $\mu + \sigma$).
 - 95% lie within two standard deviations ($\mu - 2\sigma$ to $\mu + 2\sigma$).
 - 99.7% lie within three standard deviations ($\mu - 3\sigma$ to $\mu + 3\sigma$).



D. Summary

To summarize:

- **Continuous Random Variable:** A variable capable of assuming any value within an interval, providing an unbroken spectrum of possible outcomes. Examples include height, weight, time, and temperature. *Diagram Reminder:* Number line showing the continuum of values.
- **Continuous Probability Distribution:** Defined by a probability density function (pdf), where individual points have zero probability and only intervals (areas under the curve) have meaningful probabilities. The total area under the pdf equals 1. *Diagram Reminder:* Shaded area under a generic pdf.
- **Normal Distribution:** A special continuous distribution that is bell-shaped and symmetric. Its characteristics are entirely determined by its mean (μ) and variance (σ^2).
 - **Key properties include:**
 - **Symmetry:** Mean, median, and mode coincide.
 - **Total probability (area) is 1.**
 - **The tails are asymptotic to the x-axis.**
 - **Inflection points occur at $\mu \pm \sigma$.**
 - **Empirical Rule (68-95-99.7):** Approximately 68% of values fall within one standard deviation, 95% within two, and 99.7% within three standard deviations. *Diagram Reminders:*
 - A clearly labeled bell curve showing the center, inflection points, and shaded empirical rule intervals.
 - Side-by-side diagrams displaying the effect of different parameters.

CHAPTER 7.2: USING TABLES TO FIND PROBABILITIES OF THE STANDARD NORMAL DISTRIBUTION Z

(Reference: [3, p. 155–156])

In this section, we explain the standard normal distribution and illustrate how to use its corresponding tables to find cumulative probabilities. We also detail the process of standardizing any normally distributed variable. The ultimate goal is to empower you to quickly compute probabilities for any normal distribution by referencing a single, standard table.

1. Standard Normal Distribution

Definition and Key Attributes

- **Special Case:** The standard normal distribution is a specific form of the normal distribution defined by a mean $\mu = 0$ and a standard deviation $\sigma = 1$ (thus the variance $\sigma^2 = 1$).

- **Notation:**

$$Z \sim N(0,1)$$

- **Purpose:** By standardizing any normal variable to have these parameters, we can universally apply one table—the standard normal table—to our probability calculations.

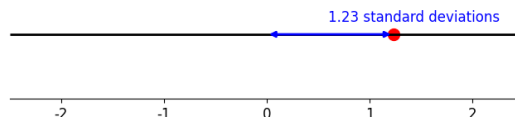
Diagram Suggestion 1

- **Location:** Immediately after this explanation
- **Content:** A bell-shaped curve labeled “Standard Normal Distribution” with the horizontal axis marked $Z = 0$ at the center. Use decorations such as:
 - A vertical line through $Z = 0$ (the mean).
 - Arrows displaying the symmetric nature of the curve on either side.
 - Notations stating “ $\mu = 0$ ” at the center and “ $\sigma = 1$ ” along the horizontal axis.

2. Standard Normal Variable (Z)

Definition and Explanation

- **Standard Normal Variable:** Any value drawn from the standard normal distribution is called a **z-score** or standard normal variable.
 - **Interpretation:** The z-score tells you how many standard deviations a particular value is from the mean. For example, $Z = 1.23$ means that the value is 1.23 standard deviations above the mean of 0.
- **Example:** Consider a data point with a z-score of 1.23. This indicates that this value lies 1.23 standard deviations to the right of the mean.



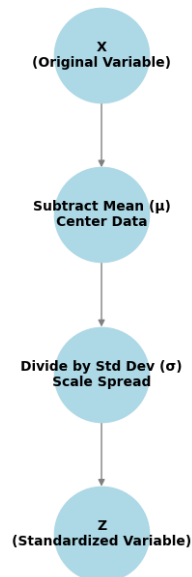
3. Standardizing a Normal Variable (X)

The Process of Standardization

- **Objective:** Convert a normally distributed variable X (with mean μ and standard deviation σ) into a standard normal variable Z . This transformation allows you to use the standard normal table for any normal distribution.
- **Formula:**

$$Z = \frac{X - \mu}{\sigma}$$

- **Step-by-Step Example:** Suppose X represents exam scores where $\mu = 70$ and $\sigma = 10$. If a student scores $X = 85$:
 1. Subtract the mean: $85 - 70 = 15$
 2. Divide by the standard deviation: $\frac{15}{10} = 1.5$. Thus, the standardized score is $Z = 1.5$.



4. Standard Normal Distribution Tables

What They Are and How They Work

- **Definition:** Standard normal distribution tables are pre-calculated tables that list the cumulative probabilities $P(Z < z)$ for various z -values. The entry in the table gives the area under the standard normal curve to the left of that z -value.
- **Structure of the Table:**
 - **Rows:** Represent the integer and the first decimal part of the z -value (e.g., 0.0, 0.1, 0.2, etc.).
 - **Columns:** Represent the second decimal digit (from 0.00 up to 0.09).
- **Usage:** To look up $P(Z < z)$, split z into its two parts (e.g., for $z = 1.23$, use row "1.2" and column "0.03") and find the intersection in the table.

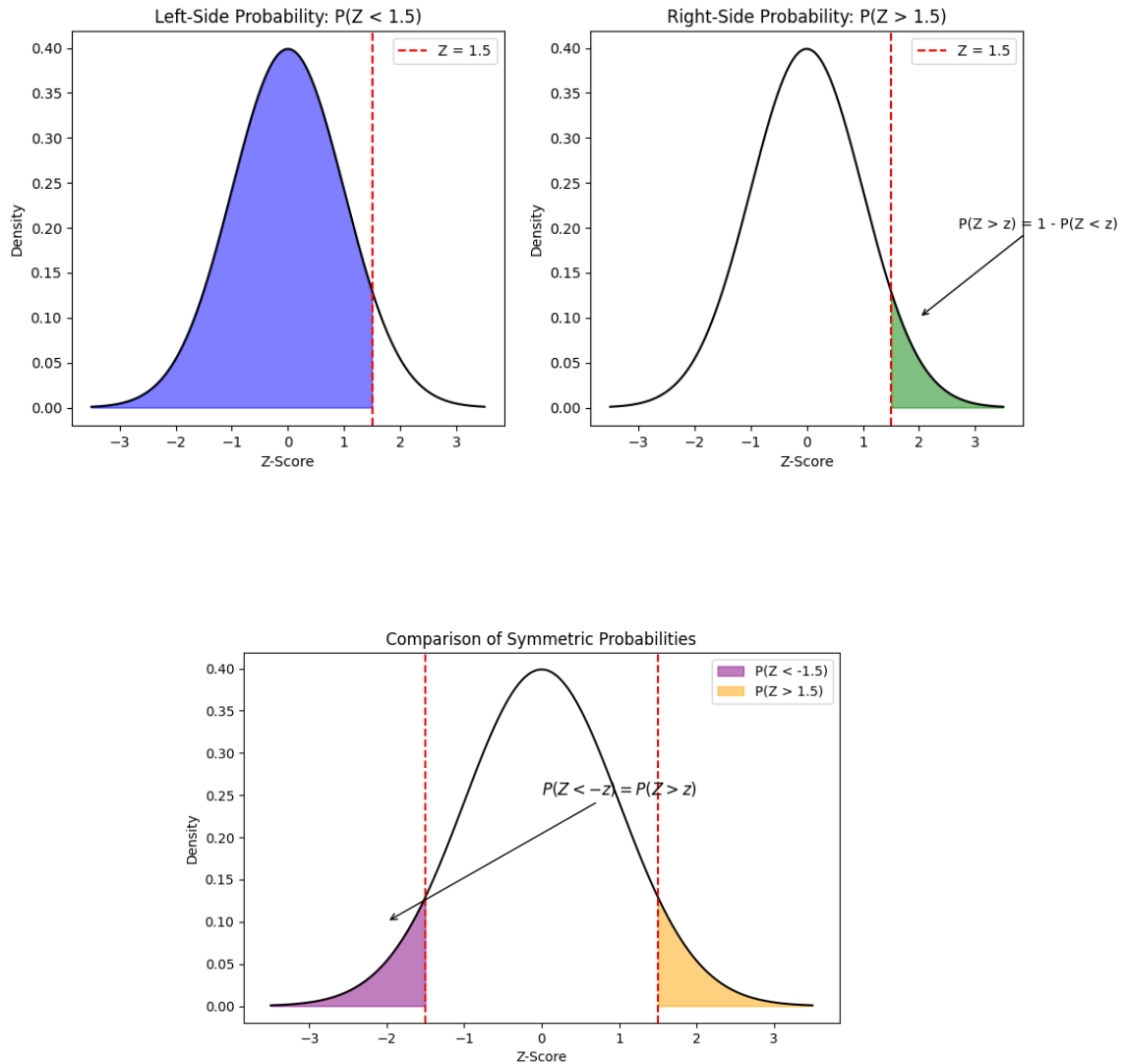
5. Using the Tables to Find $P(Z < z)$

Step-by-Step Procedure

1. **Break Down the z -value:** For a given z (say, $z = 1.23$):
 - Identify the row for the integer part and first decimal: here, 1.2.
 - Identify the column for the second decimal: here, 0.03.
2. **Locate the Probability:** The cell where the row and column intersect gives $P(Z < 1.23)$.
 - **Example:** Suppose the table indicates 0.8907 at this intersection. Then,

$$P(Z < 1.23) \approx 0.8907$$

6. Using Symmetry Properties to Find Other Probabilities



Understanding and Applying Symmetry

The standard normal distribution's symmetry around $Z=0$ allows for simple relationships to compute probabilities for values not directly listed in the standard table.

- **Right-Tail Probability:**

$$P(Z > z) = 1 - P(Z < z)$$

- **Example:** If $P(Z < 1.23) = 0.8907$, then:

$$P(Z > 1.23) = 1 - 0.8907 = 0.1093$$

- **Left-Tail for Negative z-values:**

$$P(Z < -z) = 1 - P(Z < z)$$

- **Example:** $P(Z < -1.23) = 1 - P(Z < 1.23) = 1 - 0.8907 = 0.1093$ Alternatively, because of symmetry, $P(Z < -1.23)$ equals $P(Z > 1.23)$.
- **Complement for Negative Values:**

$$P(Z > -z) = P(Z < z)$$

This equality holds because the area to the left of a positive z equals the area to the right of its corresponding negative z .

- **Probability Between Two Values:** To find the probability that Z lies between two values a and b , use:

$$P(a < Z < b) = P(Z < b) - P(Z < a)$$

- **Example:** For $0.50 < Z < 1.23$, if the table gives $P(Z < 1.23)$ and $P(Z < 0.50)$, then subtract the latter from the former to obtain the probability between these limits.

Summary

In summary:

- **Standard Normal Distribution:** A fundamental distribution defined by $\mu = 0$ and $\sigma = 1$; it simplifies calculations by providing a universal scale.
- **Standard Normal Variables (Z-scores):** Every value of Z represents the number of standard deviations it is away from the mean. This allows for direct comparisons across different datasets.
- **Standardization Process:** Any normal variable X with mean μ and standard deviation σ can be converted to the standard normal form using:

$$Z = \frac{X - \mu}{\sigma}$$

This transformation is key to using the standard normal table for probability calculations.

- **Using the Standard Normal Table:** The tables provide cumulative probabilities $P(Z < z)$. By splitting a z -value into its two components (rows and columns), we can quickly find the corresponding area under the curve.
- **Symmetry Properties:** Due to the symmetric nature of the standard normal distribution:
 - $P(Z > z) = 1 - P(Z < z)$
 - $P(Z < -z) = P(Z > z)$
 - The probability between two values is determined by the difference $P(Z < b) - P(Z < a)$.

CHAPTER 7.3: USING TABLES TO FIND THE VALUE OF z GIVEN A PROBABILITY

(Reference: [3, p. 158])

In many statistical problems, you are not given a z -value but rather a probability. The goal in such cases is to find the z -value that corresponds to that cumulative probability. This process is essentially the reverse of using the z -table to find a probability from a given z -value and is foundational in hypothesis testing and confidence interval estimation.

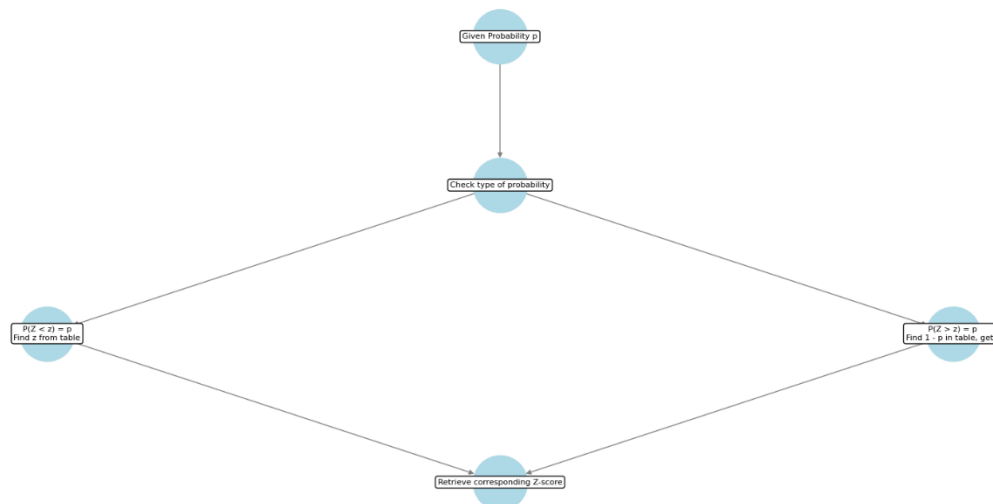
1. Overview and Fundamental Concepts

1.1 The Reverse Lookup Problem

- **Concept:** Instead of determining the probability $P(Z < z)$ for a given z -score, you are asked to find the z -score when you are given a probability.
- **Example Situation:** Imagine you're told that 89.07% of observations fall below a certain value in a standard normal distribution. You must determine the z -value, call it z^* , such that:

$$P(Z < z^*) = 0.8907$$

- **Purpose:** This reverse lookup is essential when:
 - Setting critical values for hypothesis tests.
 - Constructing confidence intervals.
 - Converting confidence levels to z -scores.



2. Finding z Given $P(Z < z) = p$

2.1 Step-by-Step Procedure

When presented with a left-tail probability p , the process is as follows:

1. **Locate the Probability in the Table:** Search within the standard normal distribution table for the probability value that is equal to (or very near) p .
 - o **Tip:** The table is divided into rows and columns. Rows represent the z -value rounded to the first decimal (e.g., 1.2) and columns represent the second decimal portion (e.g., 0.03).
2. **Read off the z -value:** Once you find the probability closest to p , note the corresponding z -value by combining the row header (integer and first decimal) and the column header (second decimal).
 - o **Example:** Suppose you are given $P(Z < z) = 0.8907$. You then locate 0.8907 in the table, which might appear at the intersection of the row labeled "1.2" and the column labeled "0.03". Thus, the corresponding z -value is:

$$z = 1.23$$

2.2 In-Depth Example

- **Scenario:** Assume you are provided with $P(Z < z) = 0.9750$.
- **Process:**
 1. Scan the body of the z -table to locate the value 0.9750 or the closest approximation.
 2. You find that 0.9750 corresponds to the intersection of the row "1.9" and the column "0.05."
- **Result:** Thus, the z -value is:

$$z = 1.95$$

- **Interpretation:** This tells you that 97.50% of the data in a standard normal distribution lie below a z -score of 1.95.

3. Finding z for Upper-Tail Probabilities $P(Z > z) = p$

3.1 Understanding the Complement Rule

- **Concept:** The standard normal table is typically arranged to show cumulative probabilities from the left tail. When you are given a probability in the upper tail, you must convert it to the corresponding left-tail probability.
- **Complement Rule:**

$$P(Z < z) = 1 - P(Z > z)$$

- **Procedure:**
 1. **Given:** $P(Z > z) = p$
 2. **Compute:** $P(Z < z) = 1 - p$
 3. Use the standard normal table to find the z -value that corresponds to $1 - p$.

3.2 Example

- **Scenario:** Suppose you are told that $P(Z > z) = 0.1093$.
- **Steps:**

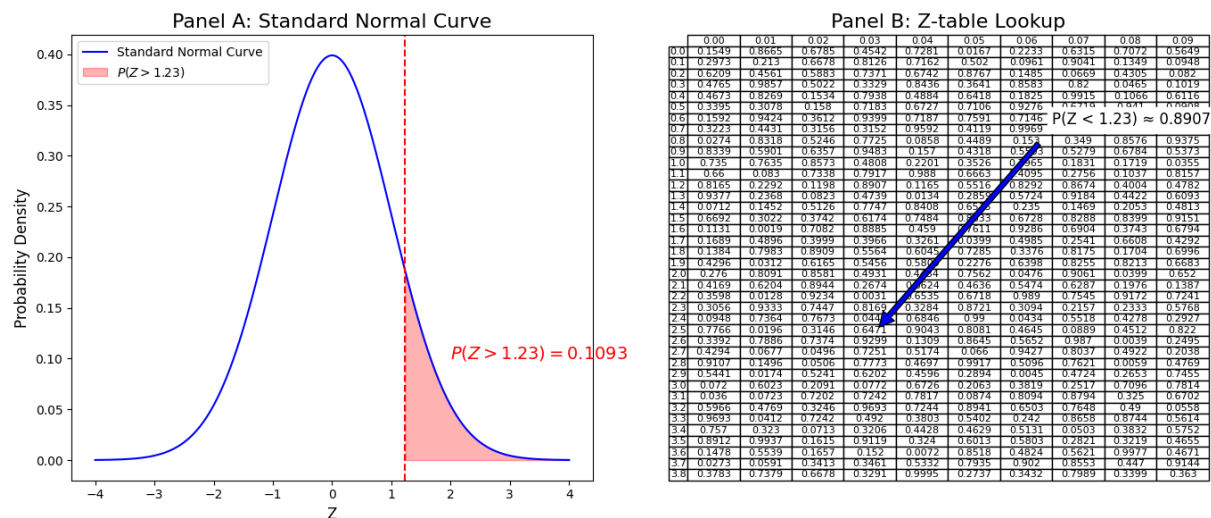
1. **Convert to Left-Tail Probability:**

$$P(Z < z) = 1 - 0.1093 = 0.8907$$

2. **Lookup:** Use the table as outlined in Section 2 to find the z-value for 0.8907, which as previously seen is approximately 1.23.

- **Result:** Thus, the corresponding z-value is:

$$z = 1.23$$



4. Using Symmetry for Negative z-Values

4.1 The Role of Symmetry in the Standard Normal Distribution

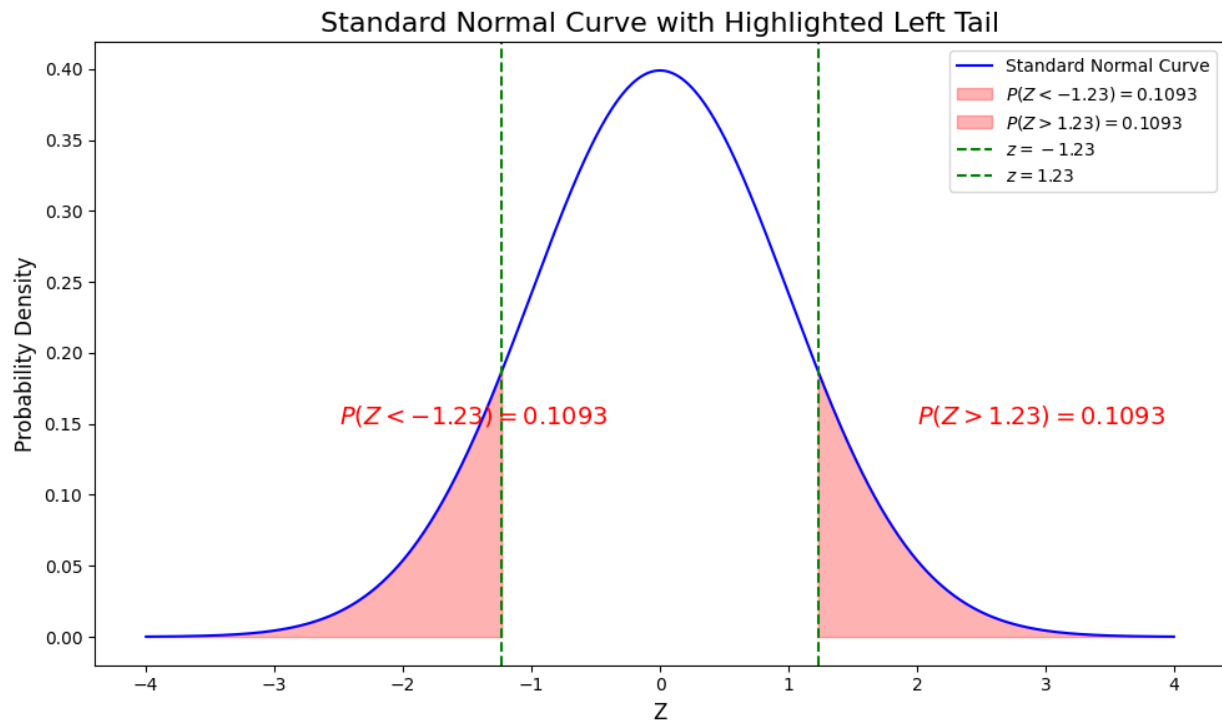
- **Symmetry Principle:** The standard normal distribution is perfectly symmetric about zero, meaning the area to the left of a negative z-value is the same as the area to the right of the corresponding positive z-value.
- **Application:** If a given probability pp is less than 0.5, this indicates that the corresponding z-value will be negative.

4.2 Example and Process

- **Scenario:** Find the z-value when $P(Z < z) = 0.1093$.
- **Procedure Using Symmetry:**
 1. **Recognize:** Since 0.1093 is less than 0.5, the z-value will be negative.

2. **Find the Positive Equivalent:** Look up $P(Z < z')$ where $z' = 1 - 0.1093$ or equivalently use the symmetry property: $P(Z < -z) = P(Z > z)$. From earlier, we saw that $P(Z > 1.23) \approx 1 - 0.8907 = 0.1093$.
3. **Assign the Negative Sign:** Therefore, the z-value corresponding to $P(Z < z) = 0.1093$ is:

$$z = -1.23$$



5. Summary

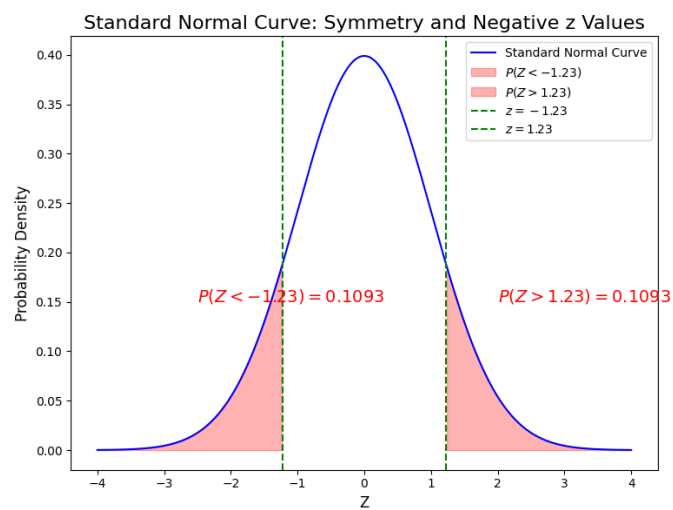
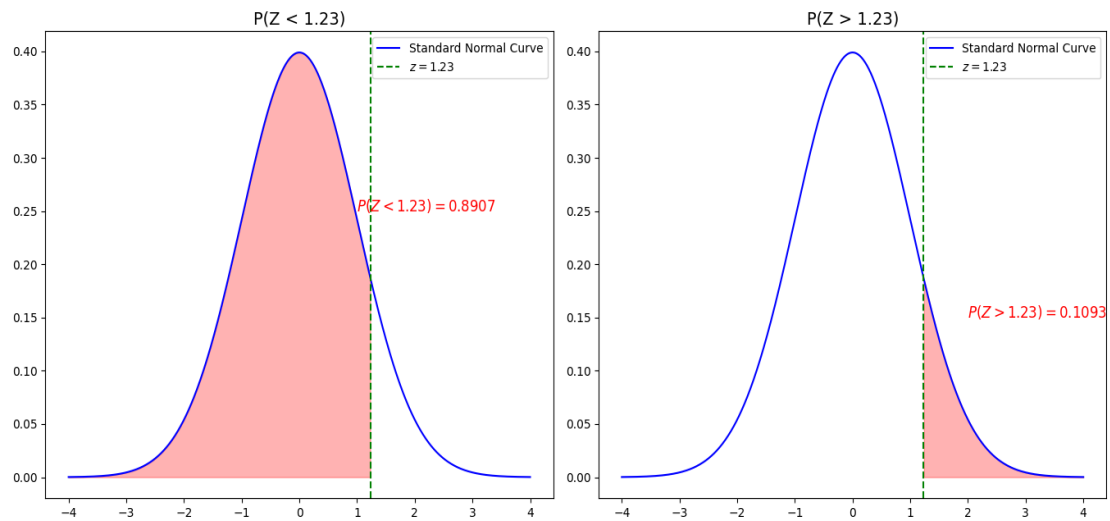
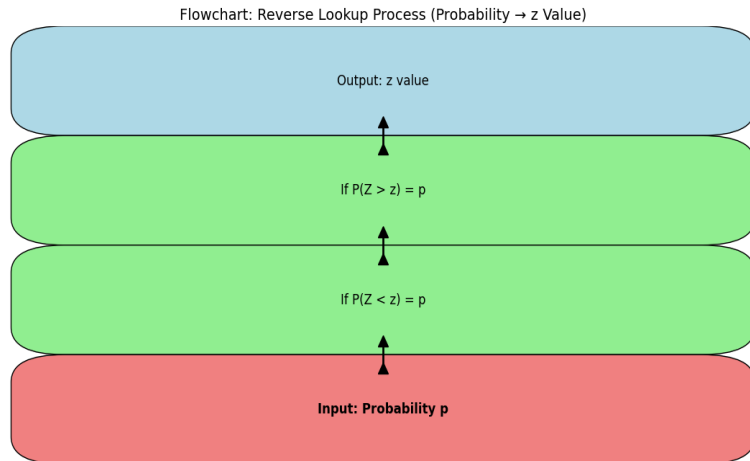
Key Points Recap

- **Reverse Lookup:** When provided with a probability, the goal is to work backward through the standard normal table to find the z-value that defines that cumulative area.
- **Steps for Left-Tail Probability $P(Z < z) = p$:**
 1. Search the standard normal table for the value closest to p .
 2. Read off the z-value at the corresponding row and column.
- **Upper-Tail Probabilities $P(Z > z) = p$:** Convert the right-tail probability to a left-tail probability using:

$$P(Z < z) = 1 - p$$

Then proceed with the standard lookup.

- **Symmetry for Negative Values:** If $p < 0.5$, the z-value is negative; you can find the corresponding positive z-value using the table, then assign a negative sign.



CHAPTER 7.4: THE STANDARD NORMAL DISTRIBUTION

(Reference: [3, p. 161])

This section reinforces the fundamental role of the standard normal distribution as a universal benchmark for analyzing any normally distributed variable. By converting any normal variable into its z-score, we can use one single set of cumulative probability tables to compute probabilities, regardless of the variable's original mean or variance.

1. Concept and Significance of the Standard Normal Distribution

Definition and Overview

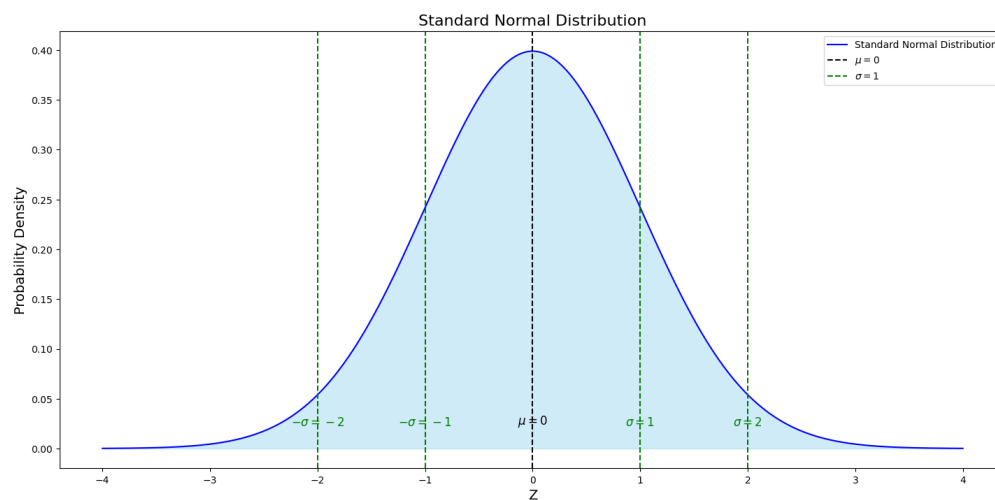
- **Standard Normal Distribution:** This is a special case of the normal distribution defined by a mean, $\mu = 0$, and a standard deviation, $\sigma = 1$ (thus the variance $\sigma^2 = 1$). It is denoted as:

$Z \sim N(0, 1)$

- **Purpose:** The primary reason for standardizing a normal variable is to convert various normal distributions (with different means and standard deviations) into one common framework. This process simplifies probability calculations since every normal variable can be transformed into the standard normal form.
- **Key Idea:** Any normally distributed variable $X \sim N(\mu, \sigma^2)$ can be transformed to a standard normal variable Z using the formula:

$$Z = \frac{X - \mu}{\sigma}$$

This transformation linearizes the variable onto a universal scale, allowing the use of standard z-tables.



2. The Process of Standardization

Step-by-Step Explanation

- **The Transformation Formula:** To standardize any normal variable X , use:

$$Z = \frac{X - \mu}{\sigma}$$

Here, subtracting the mean centers the distribution about 0, and dividing by the standard deviation scales the data so that its spread is measured in units of standard deviations.

- **Detailed Example:**

Example 1: Suppose exam scores are normally distributed such that $X \sim N(70, 10^2)$. If a student scores 85, what is the corresponding z-score?

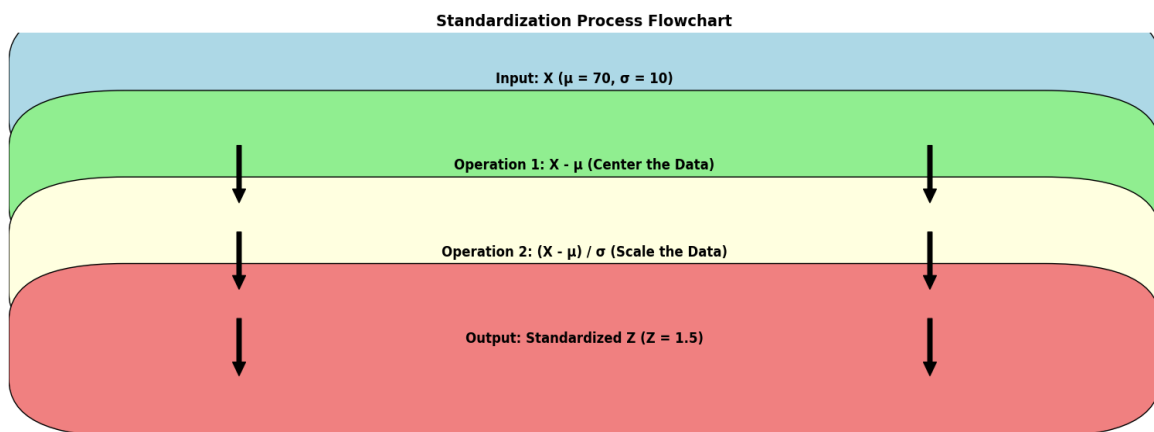
- **Step 1:** Subtract the mean:

$$85 - 70 = 15$$

- **Step 2:** Divide by the standard deviation:

$$Z = \frac{15}{10} = 1.5$$

Interpretation: A score of 85 is 1.5 standard deviations above the mean.



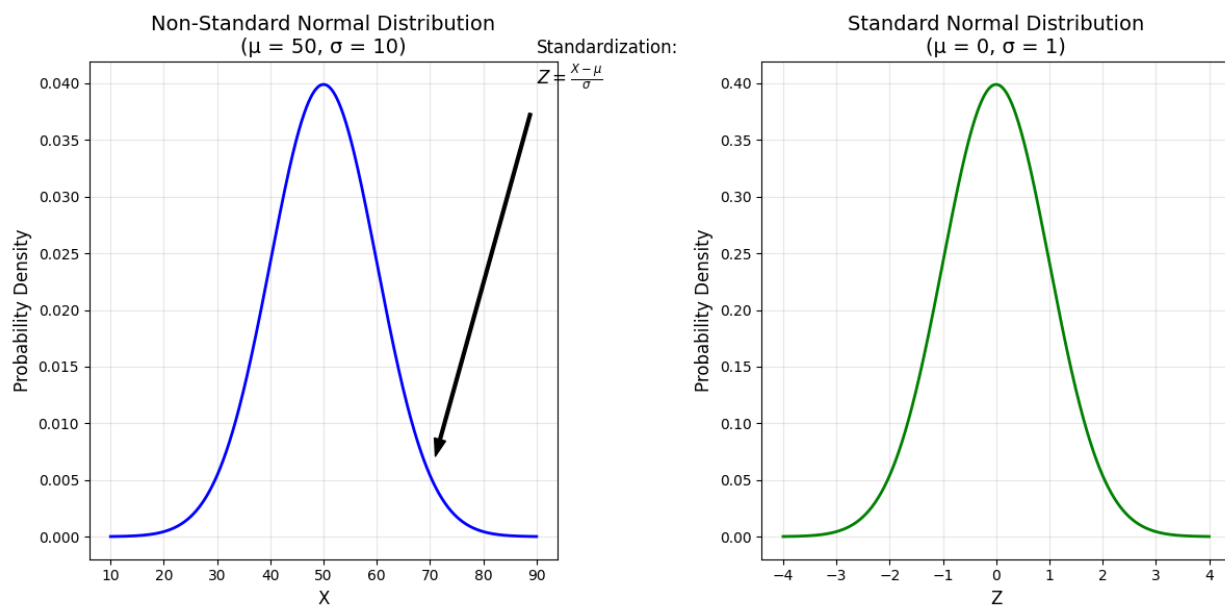
3. Using the Standard Normal Distribution Table

Standardization in Practice

- **How It Works:** Once a variable is standardized, you use a single table to find cumulative probabilities $P(Z < z)$ for any z-score.

- **Example:** With $Z = 1.5$ obtained from the previous standardization example, one can look up $P(Z < 1.5)$ in the z-table, which might yield approximately 0.9332. This tells you that 93.32% of the distribution lies below that score.
- **Interpretation and Utility:** The ability to convert to the standard normal form makes it possible to determine probabilities, set critical values, and perform statistical tests using a common reference.

	0.00	0.01	0.02	0.03	0.04	0.05
1.0	0.8413	0.8438	0.8461	0.8484	0.8508	0.8531
1.1	0.8643	0.867	0.8693	0.8717	0.874	0.8763
1.2	0.8849	0.8879	0.8902	0.8925	0.8948	0.8971
1.3	0.9332	0.9336	0.9352	0.9365	0.9378	0.9391



5. Summary

Key Points Recap

- **Definition:** The standard normal distribution, $Z \sim N(0,1)$, is a normal distribution with a mean of 0 and a standard deviation of 1, serving as a foundational reference for all normal distributions.
- **Standardization Process:** Any normal variable $X \sim N(\mu, \sigma^2)$ can be transformed into a standard normal variable using:

$$Z = \frac{X - \mu}{\sigma}$$

This process centers and scales the data, making it compatible with standard z-tables.

Diagram Recaps

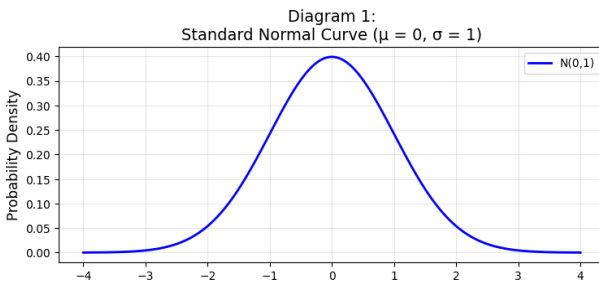
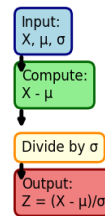


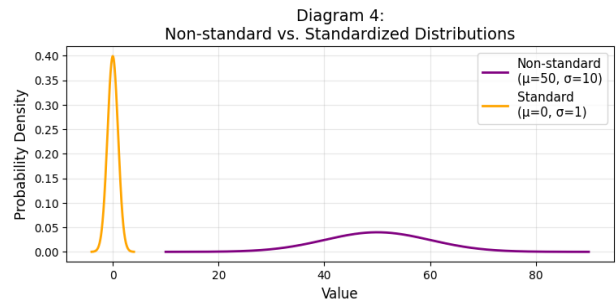
Diagram 3:
Snapshot of Standard Normal Table

Z	0.00	0.01	0.02	0.03
0.0	0.5000	0.5040	0.5080	0.5120
0.1	0.5398	0.5438	0.5478	0.5517
0.2	0.5793	0.5832	0.5871	0.5910
0.3	0.6179	0.6217	0.6255	0.6293

Diagram 2:
Standardization Flowchart



Example:
 $X = 85, \mu = 70, \sigma = 10$
 $\Rightarrow X - \mu = 15$
 $\Rightarrow Z = 15/10 = 1.5$



CHAPTER 7.5: FINDING μ AND σ

(Reference: [3, p. 164])

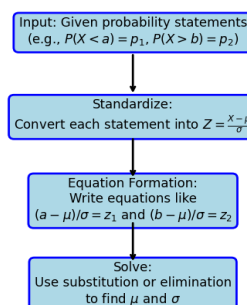
In many real-world applications, the parameters of a normal distribution—the mean μ and the standard deviation σ —are not given directly. Instead, you are often provided with probability information (e.g., cumulative probability statements) that can be used to determine these parameters. In this chapter, we detail a systematic approach to translate probability statements into equations, and then solve for μ and σ .

1. Overview and Concept

1.1 Context and Importance

- Objective:** When μ and/or σ are unknown, available probability statements about the variable X (which is assumed to be normally distributed) are used to create one or more equations. These equations are then solved for the missing parameters.

Diagram 1: Overview Flowchart



2. Translating Probability Information into Equations

2.1 Basic Principle

- **General Approach:**
 - When given a probability statement involving X such as:

$$P(X < x_0) = p$$

you first standardize X using:

$$Z = \frac{X - \mu}{\sigma}.$$

- The probability statement becomes:

$$P\left(Z < \frac{x_0 - \mu}{\sigma}\right) = p.$$

- **Finding the Equivalent z-Value:** Use the standard normal distribution table to find the z-value (z_p) that corresponds to the cumulative probability p . Then you can set up the equation:

$$\frac{x_0 - \mu}{\sigma} = z_p.$$

2.2 Detailed Example

- **Example 1:** Suppose you know that

$$P(X < 50) = 0.8413,$$

and X is normally distributed with unknown μ and σ .

1. **Standardize the Statement:**

$$P\left(Z < \frac{50 - \mu}{\sigma}\right) = 0.8413.$$

2. **Find z-value:** The standard normal table tells us that

$$P(Z < 1) \approx 0.8413$$

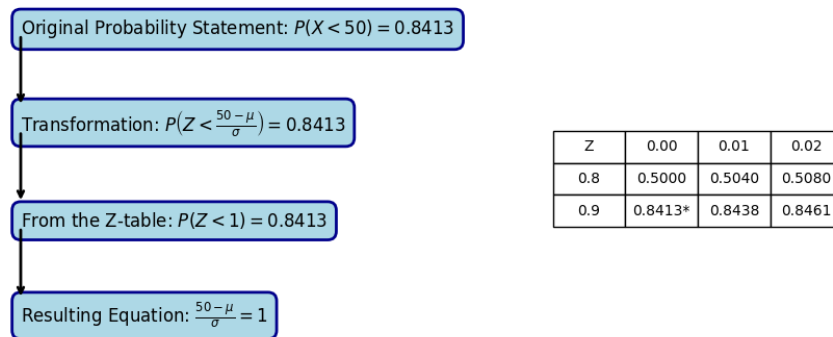
3. **Set Up the Equation:** Therefore, we obtain:

$$\frac{50 - \mu}{\sigma} = 1,$$

which can be rearranged to:

$$50 - \mu = \sigma.$$

Diagram 2: Equation Transformation for Standardization



3. Solving for μ and σ

3.1 Single Probability Statement (One Parameter Unknown)

- **Scenario:** If either μ or σ is known, one probability statement provides enough information to solve for the unknown parameter.
 - **Example:** If σ is known, then from:

$$\frac{50 - \mu}{\sigma} = 1,$$

we solve for μ as:

$$\mu = 50 - \sigma.$$

3.2 Two Probability Statements (Both Parameters Unknown)

- **General Process:** When both μ and σ are unknown, at least two independent probability statements are needed to solve the system.
 - **Example:** Suppose we are given:
 1. $P(X < 10) = 0.2$
 2. $P(X > 15) = 0.1$

Step-by-Step Example

1. **First Probability Statement:** $P(X < 10) = 0.2$
 - **Standardize:**

$$P\left(Z < \frac{10 - \mu}{\sigma}\right) = 0.2.$$

- **Using the z-table:** For $p = 0.2$, the corresponding z-value is approximately $z \approx -0.84$ (since $P(Z < -0.84) \approx 0.2$).
- **Equation 1:**

$$\frac{10 - \mu}{\sigma} = -0.84$$

2. **Second Probability Statement:** $P(X > 15) = 0.1$
 - **Convert:** Since $P(X > 15) = 0.1$, then

$$P(X < 15) = 1 - 0.1 = 0.9.$$

- **Standardize:**

$$P\left(Z < \frac{15 - \mu}{\sigma}\right) = 0.9.$$

- **From the z-table:** The z-value for 0.9 is approximately $z \approx 1.28$.
- **Equation 2:**

$$\frac{15 - \mu}{\sigma} = 1.28.$$

3. **System of Equations:** We now have:
 - Equation (1): $\frac{10 - \mu}{\sigma} = -0.84 \Rightarrow 10 - \mu = -0.84 \sigma$.
 - Equation (2): $\frac{15 - \mu}{\sigma} = 1.28 \Rightarrow 15 - \mu = 1.28 \sigma$.
4. **Solving the System:** Subtract Equation (1) from Equation (2):

$$\begin{aligned} (15 - \mu) - (10 - \mu) &= 1.28\sigma - (-0.84\sigma) \\ 15 - 10 &= 1.28\sigma + 0.84\sigma \\ 5 &= 2.12\sigma \Rightarrow \sigma = \frac{5}{2.12} \approx 2.36. \end{aligned}$$

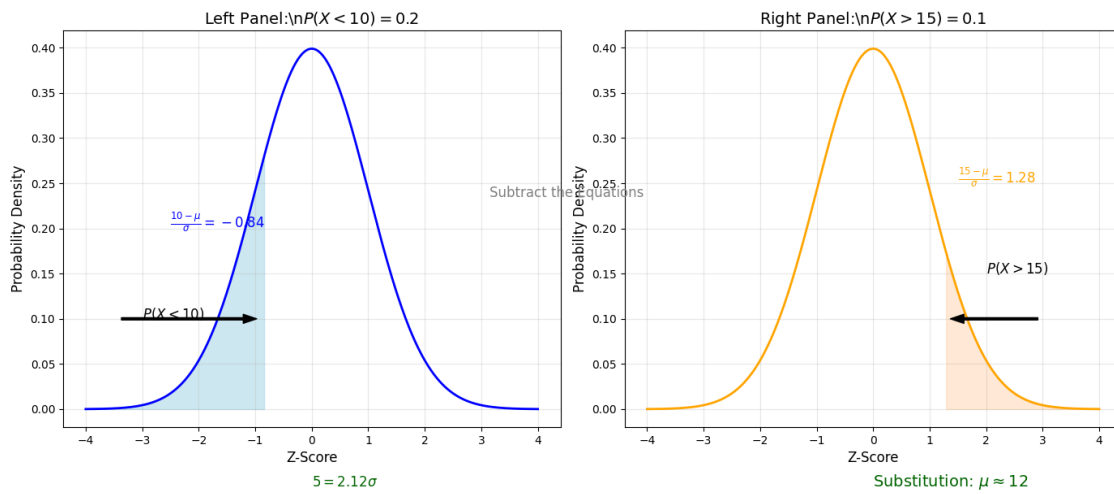
Substitute $\sigma \approx 2.36$ back into Equation (1) (or Equation (2)) to solve for μ :

$$10 - \mu = -0.84 \times 2.36 \approx -1.98 \Rightarrow \mu = 10 + 1.98 \approx 11.98.$$

5. **Result:** The approximate values are:

$$\mu \approx 12 \quad \text{and} \quad \sigma \approx 2.36.$$

Diagram 3: System of Equations for Solving μ and σ



4. Summary

Key Points Recap

- **Purpose:** To determine the unknown mean μ and standard deviation σ of a normal distribution using given probability statements.
- **General Approach:**
 1. **Translate Probability Statements:** Convert statements like $P(X < x_0) = p$ into standardized form:

$$P\left(Z < \frac{x_0 - \mu}{\sigma}\right) = p.$$

2. **Find Corresponding z-Values:** Use the standard normal distribution table to find the z-value z_p associated with p.
3. **Set Up Equations:** Write equations such as:

$$\frac{x_0 - \mu}{\sigma} = z_p.$$

4. **Solve the Equations:** If one parameter is unknown, directly solve the equation. If both μ and σ are unknown, use at least two independent probability statements to form a system of equations and solve by substitution or elimination.
- **Detailed Example Recap:** Using two probability statements— $P(X < 10) = 0.2$ and $P(X > 15) = 0.1$ —we derived a system of equations:

$$\frac{10 - \mu}{\sigma} = -0.84 \quad \text{and} \quad \frac{15 - \mu}{\sigma} = 1.28.$$

Solving these yielded approximate values $\mu \approx 12$ and $\sigma \approx 2.36$.

Diagram 1: Overall Flowchart

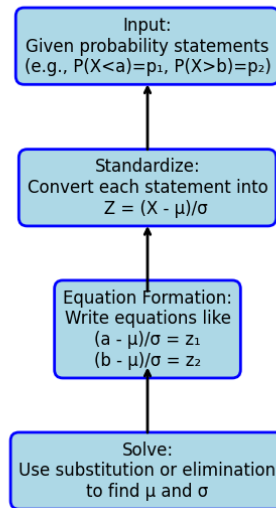


Diagram 2: Transformation Diagram

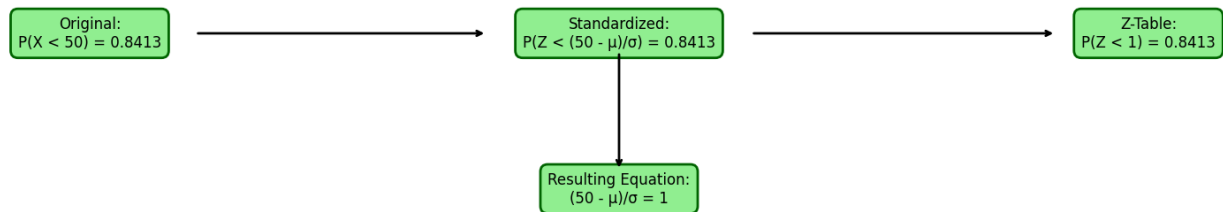


Diagram 3: System of Equations Diagram

